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Report on the R&D project:

Development of resilient production systems for biodynamic bananas in accordance with the organism principle (BD Banana)

Phase 1 – Preliminary tests for system optimization

Project management

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Project

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Project duration:

Planned: 04/2024-03/2025 (12 months)

Extended: 04/2024 – 09/2025 (18 months)

Project objective and approach

Objective

The objective was to work with Demeter-certified farms in the Dominican Republic to further develop the ecological potential of biodynamic banana cultivation based on the concept of organisms, thereby increasing the ecological and economic sustainability of production. The results should also contribute to the sustainable transformation of banana cultivation beyond biodynamic farming.

The development of farm organisms using agroecological methods was intended to increase productivity while reducing external costs and achieving positive ecological and socio-economic side effects.

Approach

As part of the project, various agroecological measures were tested and analyzed in collaboration with 10 certified Demeter farms in the Dominican Republic covering more than 1,200 hectares.

The project focused on increasing biological N fixation and N efficiency within the agricultural organism in order to increase farm productivity and independence from external inputs, especially organic fertilizers.

We also expected positive effects on other areas of the farm organism:

- The integration of (more) animals is intended to expand the source of income from local sales to include animal products.
- The integration of (more) trees and more plant species (including flowering plants) should promote biodiversity on the farm
- Improving soil fertility should have a positive impact on soil performance (e.g., water storage and water use efficiency).

The tested measures were evaluated in terms of work efficiency, water use, and crop diversity and initially integrated into farm practice.

Background/problem

The guiding principle of biodynamic farming is to create as closed an agricultural organism as possible. For this reason, the use of external inputs is much more restricted than in other areas of organic farming. Bananas are a crop that requires very heavy fertilization (in conventional practice, approx. 600 kg N/year) in order to achieve the economically necessary yields (more than 40 t/ha/year). For this reason, the farms have applied for an exemption to be able to use more commercial organic fertilizers (120 kg N from commercial fertilizers instead of the 40 kg N permitted in the guidelines).

This was approved on condition that the farms attempt to mobilize more nitrogen sources within their own operations and increase nitrogen efficiency.

This led to the development of this practical research approach, which examines which operational measures are suitable for increasing nitrogen fixation across the entire farm and thereby achieving the necessary yield level while simultaneously reducing costs.

Project implementation and results

The aim was to test various strategies for optimizing production ecology in biodynamic banana cultivation. To this end, a practical research network was established in collaboration with the farms in order to develop an understanding of the framework conditions and options for action and their ecological, economic, and social implications. A total of 11 demonstration trials were carried out and evaluated in the practical research network on the participating farms.

The project built up expertise in conducting practical trials and trained the partners accordingly. On this basis, a number of scientific trials have already been conducted or set up. Further trials and optimization approaches were developed jointly in a follow-up project phase.

The following six measures were agreed upon:

1. Practical research network

The network comprises four companies with a total of ten farms:

Plantaciones del Norte is owned by an Italian family. The two managing directors, Andrea Bolini and Marino Trastullio, are co-partners. Plantaciones has seven farms of its own, six banana farms with a total of 590 hectares of banana cultivation area and one lemon farm. All farms are Demeter-certified, and some are also certified according to the Fairtrade standard. In addition, they are also exporters of organic and Fairtrade bananas and purchase from certified organic and Fairtrade farms. The switch to Demeter was made for marketing reasons.

Savid. The owner and managing director is Jetta van den Berg, a successful pioneer of organic banana cultivation in the Dominican Republic for 30 years. She owns two farms with 270 hectares, her own dairy cattle and beef cattle, and a flock of sheep. Solar energy is used to operate the water pumps for irrigation. The farms are certified by Fairtrade and Biosuisse. Savid mainly markets its products to COOP in Switzerland. The switch to Demeter was initially made at the request of COOP. However, the owner is now convinced of the concept.

Probanor is a young farm with 100 hectares. The owner is Karl Steiner, a Swiss doctor. The farm managers are two young Dominican engineers who have leased the farm from Karl Steiner and aim to develop it into a model farm. The farm is Fairtrade certified. It is in the process of establishing its own herd of cattle (currently 17 cows). Karl Steiner's mother practiced biodynamic farming in her garden, and the two farm managers are convinced of the concept.

Biotropic is the longest Demeter-certified farm with two parts totaling 120 hectares. Marketing is mainly carried out in the German natural food trade.

A database accessible to all was set up, in which all trials were documented with trial protocols and pictures. For sensitive data, an area accessible only to the respective farm was set up.

Regular joint farm visits took place, as well as regular coordination meetings with exchanges of experience and discussion.

2. Trial experiments

Producers usually learn by observing successful measures. Therefore, the main task of the exploratory trials was to show producers measures that seemed sensible and to develop further questions from these. Measures that were accepted were to be integrated into farm practice as soon as possible.

2.1. Trial cultivation of legumes

Methodology:

All legumes for which seeds were available were cultivated on the Probanor farm on an area of 0.5 ha. The area is irrigated by flooding. The aim was to observe germination behavior, growth rate, and weed competitiveness.

Results:

The high weed pressure, for which there is no operational experience in controlling, made evaluation difficult. No exact data was collected, but valuable observations were made:

- Kudzu (*Pueraria montana*) did not germinate, nor did Mani foragero (*Aranchis pintoi*).
- *Alfalfa* developed very well but disappeared in the second growth: probably due to flood irrigation, which alfalfa does not tolerate well in its early stages.
- *Crotalaria juncea* and *Canavalia ensiformis* impressed with their rapid growth and good competitiveness with weeds. Caupi (*vigna unguiculata*) also grows rapidly but is less competitive.
- *Clitoria ternatea* has slow early development but nevertheless holds its own well under weed pressure as it is a shade-tolerant plant.

Crotalaria with high sowing density

Different sowing densities of Crotalaria

Well-established alfalfa crop

2.2. Legume screening trial at Plantaciones, Finca Fernandez

Methodology:

Various legumes were sown in an existing banana plantation to study sowing density, biomass yield, and growth behavior.

Results:

Unsuccessful trial: sowing densities far too low; excessive weed pressure.

2.3. Further legume screening trial at Plantaciones, Finca Billy

Methodology:

This time, further sowing was carried out in a young plantation shortly after the bananas had been planted.

Results:

A lot of data on field emergence, plant density, and yield was collected. Canavalia, Crotalaria, and cowpea impressed the farm. During a tour by the expert group, it was noticeable that the young bananas had developed better in the plots with dense biomass. It seems that this protects them from the wind. Details can be found in Appendix 1.

2.4. Production of hydrolysate from Crotalaria in Plantaciones/Finca Fernandez

Methodology

Plants were harvested from the Crotalaria screening trial (approx. 276 kg) and placed in a 1000 l container and filled with water. Samples were taken and examined after 7 and 14 days.

Results:

The hypothesis that the protein in the plant decomposes and is available as N was confirmed. The majority of the protein is hydrolyzed after 7 days, but continues to break down in the second week. Details can be found in Appendix 2.

2.5. Production of hydrolysate from Crotalaria in Biotropic

Methodology:

Here, only 205 kg of plant mass was placed in the 1000 l container.

Results:

Due to the low mass and the associated higher dilution, the laboratory was unable to determine the N content.

2.6. Grazing trial with sheep in a banana plantation

At Quinta Pasadena, one of the two Savid farms, a grazing trial with sheep was conducted. It was observed that some sheep ate the young banana shoots. Therefore, there was little willingness to conduct further trials. In addition, there was concern that grazing sheep in the plantations could cause problems with Global GAP certification.

After several months there is now a protocol for grazimg with sheep in the plantation.

In the meantime, another trial has been started in plantain fields, which will be plowed after the next harvest and therefore the damage to the shoots is not relevant. Initial experiences are now being gathered.

2.7. Cultivation trials with Crotalaria, Canavalia, Caupi, and alfalfa in Paso Robles

Methodology:

Cultivation trials with these four legumes were conducted in a young banana plantation. The aim was to determine the optimal plant density, measure the biomass yield, and determine the N content of the biomass.

Results:

- Canavalia and Crotalaria performed well with high biomass yields and good weed tolerance.
- The Caupi seed was not good. However, this plant species also promises good biomass formation in a very short time. The trial must be repeated with better seed.
- For details on yields and protein content, see Appendix 3.

2.8. Large-scale cultivation trial with Crotalaria in Quinta Pasadena

Methodology:

In order to obtain a larger quantity of Crotalaria for initial hydrolysate trials at Quinta Pasadena, Crotalaria was sown on an area of 0.5 ha. Half of the area was treated with a deep loosener to investigate whether there were any visible differences in the deep rooting of the Crotalaria.

Results:

Strong nitrogen fixation was observed. Unfortunately, irrigation was inadequate, resulting in very uneven field emergence. Soil cultivation, sowing techniques, and irrigation must be significantly improved in order to achieve good results.

2.9. Transfer of the first trials with Crotalaria into operational practice

In Paso Robles, Crotalaria is now sown as green manure for all new plantings of plantains. Various variants were tested with regard to sowing density, sowing time, and different cutting heights for the first cut. Precise cultivation instructions for new plantings were developed. (Appendix 4)

2.10. Seed production trial

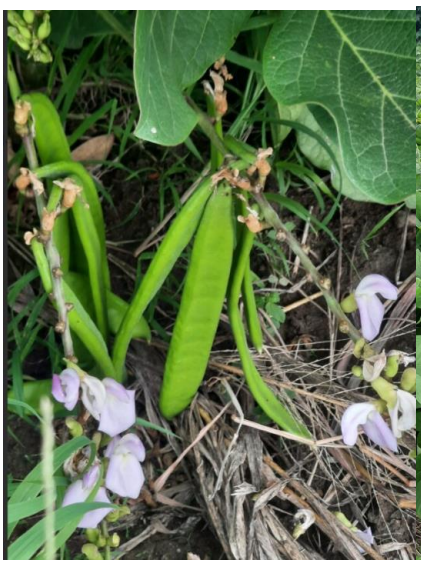
Methodology:

In Paso Robles, an experiment was set up to determine the most practical way to obtain seeds from plants that are not commercially available. The main focus is on *Mucuna*, *Canavalia*, and *Clitoria.* Pure seeds with different densities were planted on various plots.

The aim was to observe the optimal plant density and which support plants appear to be most suitable for *Mucuna* and *Clitoria*.

Results:

- *Mucuna* grows very quickly and overgrows the support plant (in this case, *Pennisetum*). The support plant should be sown earlier and the *Mucuna* later.
- *Clitoria* can be sown very densely (>10 plants/m²).
- For all three legumes, the seeds must be harvested over a longer period of time. Therefore, harvesting by hand seems to be the most sensible option.
- In a further experiment, the mechanizability of the work steps must be tested.



Canavalia ensiformis

Mucuna pruriens

Clitoria ternatea

2.11. Feeding trial of dairy cows with Crotalaria

An initial trial was conducted to determine how we can measure the effect of feeding more protein-rich fodder crops on milk yield and manure quality.

This trial must be repeated as the data obtained was not reliable.

2.12. Feed mixture cultivation trial

Methodology:

The combination of the previously cultivated *Pennisetum purpureum* with *Clitoria ternatea* was tested on a small scale. Two rows of grass and two rows of legumes were sown alternately.

Results:

- *Clitoria* can be sown even more densely (>10 plants/m²).
- *Clitoria* takes longer to develop, but also grows well in the shade of the grass
- The farm is accustomed to cutting too late. According to the literature, the optimal cutting height for balanced nutrient content is approx. 1.20 m
- The biomass share of *clitoria* in this mixture is less than 10%. According to dairy farmers, clitoria is a good milk-producing feed even in small quantities.

These experiences have now been incorporated into the next comprehensive forage cultivation trial (see 6.2.)

3. Integration of (more) ruminants

Savid already has an appropriate number of animals. On the farm, which previously only had a dairy herd, work has begun on establishing a flock of sheep to graze the plantations. Initial small-scale grazing trials with 10 animals have been so successful that another 10 animals and a ram have been added. A flock of sheep is thus being gradually established.

Probanor has a small herd of 17 cows, some of which are milked. The herd is being built up gradually from the farm's own offspring. There are plans to build up a flock of sheep.

Biotropic has a herd of beef cattle. The fodder crop trial currently underway is intended to improve the fodder base on their farm. This will enable them to keep more animals.

Plantaciones only has animals on two farms. A lot of manure is purchased. There is no willingness to increase livestock numbers here.

4. Integration of (more) trees and (more) flowering plant species (biodiversity)

After the project began, discussions quickly turned to global warming and the question of possible climate adaptation measures, which led to a discussion of agroforestry. Traditionally, banana farmers do not want trees in or around their banana fields. In this respect, there is enormous resistance to overcome.

Savid understands this, has a significant tree population, and has now created a 2.8-hectare experimental field for agroforestry. A detailed description of this approach, including all elements, can be found in a separate section below.

At **Biotropic**, work has begun on planting coconut palms in an existing plantation.

At **Probanor**, a new plantation consisting mainly of *Gliricidia* is being planted on 2 hectares, as well as some *Samane*s as shade trees.

At **Plantaciones**, a promised area has been withdrawn. It is currently unclear whether the issue of agroforestry will be pursued further.

The convincing initial trials with legumes have led to the sowing of *Crotalaria* in new plantings on three farms. The yellow-flowering *Crotalaria*, the blue-flowering *Clitoria*, and the white-flowering *Canavalia*, which are now being introduced into the plantations, enrich biodiversity.

5. Building soil fertility

This is a long-term process. The first effects are being achieved through the newly integrated legumes: better and deeper rooting of the soil. Another aspect is the planting of trees: since these are mostly legumes, the leaves and fruits also introduce additional organic matter into the farm cycle. The development of improved pasture and fodder tree mixtures leads to more organic matter in the soil on these areas.

6. Scientific field trials

6.1. Utrisha (Plantaciones and Paso Robles; completed)

A product recently introduced in Germany for use in corn cultivation that contains nitrogen-fixing bacteria that enter the leaf through the stomata and fix nitrogen and make it available to the plant, was tested for its effectiveness in banana cultivation.

6.1.1. Trial in Paso Robles

The trial conducted in Paso Robles with the agent showed no significant effects on yield. Since different fertilization levels were used in the trial, a secondary finding was that even higher fertilization had no effect. This raises the question of whether fertilization practices need to be improved. For details, see Appendix 3.

6.1.2. Trial at Plantaciones

Trials were conducted at two farms in Plantaciones. Here, too, no significant effects were observed. For details, see Appendix 4.

6.1.3. Conclusions

Parallel to the trials, a literature search was conducted and contact was made with representatives of the company that distributes the product in Central America and has also conducted its own trials. The literature contains two studies that come to positive conclusions, but also five studies that show no effects. The results provided to us by the distributor show yield increases of up to 9%.

Based on their experience, neither company intends to use the product.

6.2. Legumes and grasses for pasture and forage production

(Paso Robles and Biotropic, ongoing)

This forage production trial is an important experiment.

The aim is to develop forage and pasture mixtures that offer a higher protein content. On the one hand, this serves to improve animal performance in terms of milk and meat, and on the other hand, the protein in the feed increases the nitrogen content in the manure.

This trial is significant not only for banana farms, but for the entire agricultural sector in the Dominican Republic. Its relevance is also reflected in the interest shown by the local university in conducting final theses based on the trial. In addition, Aquilino Cruz, operations manager at Biotropic, who himself comes from a livestock farming family, has shown great interest in this topic and has made this scientific trial possible at the Biotropic farm.

<u>Methodology:</u>

We selected seven different grasses and four different legumes to test them for yield, protein content, growth rate, and grazing suitability. (Details in Appendix 5)

This is an experiment with 24 variants, each repeated three times. The experiment was conducted at both Biotropic and Paso Robles.

In addition, larger plots were sown with pasture mixtures, which will then be grazed to test their grazing suitability. Hedges of leguminous trees (Leucaena and Gliricidia) are planted between the

individual plots to serve as additional feed and reduce evaporation. These larger plots have no replicates and are used for initial observation.

Initial results:

As with other trials, soil cultivation, irrigation, and the enormous weed pressure place high demands on the implementation of such trials. In some cases, individual plots or entire replicates had to be resown.
(Aerial view of the trial, Appendix 6)

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6.3. Comparison of commercial fertilizer and self-produced fertilizer (ongoing)

Paradoxically, the nitrogen fertilizers used on organic farms originate from Europe's intensive, industrial factory farming and its slaughterhouse waste. Karl Steiner from Probanor has been experimenting with the production of a local fertilizer made from chicken feathers from a small local slaughterhouse.

Methodology:

In an experiment, these two fertilizers (Terafert and own production) are compared with two different fertilization levels and a control plot. The two fertilizers with two levels are repeated once, resulting in a total of nine plots. The plots are blinded so that the data collector is not influenced.

The following parameters are recorded:

- Speed of leaf succession
- Trunk circumference
- Number of hands on the fruit cluster
- Yield

No evaluations are available yet.

6.4. Effect of intensive horn silica application on Sigatoka infestation (ongoing)

A biodynamic banana farm in Colombia reports that intensive horn silica application has led to a reduction in Sigatoka infestation. As Sigatoka is the most economically relevant banana disease, this would significantly improve productivity.

Methodology:

Horn silica is applied monthly to an area of 4 ha using a drone, and this area is compared with an untreated control area based on selected individual plants. The last treatment was at the end of August.
An assessment of the Sigatoka infestation is carried out.

Results:

The evaluation will be available by the end of September. As the infestation pressure was very low this summer, it is doubtful whether any effect will be visible.

7. Agroforestry

In the initial phase of the trials, farm managers stated that the increasing high summer temperatures of over 40°C during several weeks lead to significant yield losses. How do we deal with this?

Since agroforestry systems are the only ones we know of that can create a different microclimate, we have tried to convince the farms to establish at least one hectare of agroforestry trial area. Only Jetta van den Berg from Savid is convinced that this is the right direction to take, and so far the first agroforestry trial has only been established in Paso Robles on an area of 2.8 hectares. At Probanor, some elements are being integrated into new plantings, but growers are very skeptical about shade trees. At Biotropic, some coconut palms have now been integrated, and shade trees may also be planted.

The system has the following elements:

7.1. Shade trees

In order to lower the temperature in the microclimate, more shade is needed, as well as more trees that contribute to evaporative cooling through transpiration. We have identified two types of trees: trees with flat crowns that create permanent shade on a specific area under their canopy. On the other hand, trees with more upright crowns tend to create a shadow that moves with the sun, so that the banana plants standing nearby are only shaded at certain times.

The trees with flat crowns should be planted on all unproductive areas and provide permanent shade. These include paths, canals, and cable car routes. A side effect of the shade trees along the cable car route is a reduced temperature under the cable car, resulting in better working conditions for the people working there.

The trees with upright crowns are to be integrated primarily into the windbreak hedges planted in a north/south direction. This gives them a triple function: shade, wind protection, and nitrogen fixation/biomass contribution.

The current favorites among shade trees are deciduous leguminous trees such as the flamboyant. When temperatures are slightly lower in winter (December to February), more sunlight is allowed into the plantation.

We prefer to plant leguminous trees because they also contribute to nitrogen supply and provide additional biomass for the mulch layer when their leaves fall.

To prevent the shade trees from using the water and nutrients of the neighboring banana plants, we have surrounded the trees with a geomembrane reaching 1 m into the ground so that the roots grow deep ( ). It remains to be seen how well this system works

Excavation with an excavator | Lining the planting hole with geomembrane | Compost

7.2. Windbreak hedges

Another element of the system is the windbreaks. If the shade trees create a cooler climate, then the cooler air should also remain in the plantation and not be constantly replaced by hot air from outside. Strong winds also cause the banana leaves to tear, reducing photosynthesis.

The windbreaks will mainly consist of two leguminous tree species:

Leucaena leucocephala (called Lino) is a very fast-growing tree that produces a lot of seeds and is therefore considered a weed. However, the young branches are excellent livestock feed. We want to integrate Lino into the windbreak hedges in such a way that it is kept short enough to prevent seed formation and the young branches are integrated into the mulch layer under the bananas.

The second tree is *Gliricidia sepium*. It is expected to grow taller and provide wind protection for the banana plants.

Individual shade trees are to be integrated into these windbreak hedges, as well as additional flowering shrubs to bring even more diversity.

Leucaena leucocephala (Lino)

Gliricidia sepium (Madre de Cacao or Pinon cubano)

7.3. Double row system

We have cultivated this agroforestry field using a double row system. In Paso Robles, this system is favored primarily for labor reasons. Our main consideration was the need to separate the banana area, which is heavily fertilized, from the legume area so that nitrogen can be effectively fixed by nodule bacteria.

The double row is approximately 1.50 m wide. The distance between the double rows is 3 to 3.5 m. Various legumes and combinations thereof are being tested in the spaces between the rows. The aim is to produce as much biomass as possible in order to build up the mulch layer under the bananas. Initially, Crotalaria is used, which receives a lot of light in the early stages of banana growth and then has the highest biomass production (up to 2 kg/m² in 45 days). Depending on the cutting height, two to three cuts are possible. Then, mainly perennial plants are cultivated, which are intended to form the permanent plant cover (*Clitoria ternatea, Mucuna pruriens, Pueraria phaseolides* (kudzu), *Neonotonia wightii* (perennial soybean), *Alfalfa tropicale*).

Since most of these plants are climbing plants, we are also experimenting with grasses, which we sow in the rows as climbing aids, as the leaf area is increased when the plants climb.

7.4. Mulch layer formation

We try to build up a sufficient layer of mulch under the double row as quickly as possible. This has three main functions: firstly, it suppresses weeds; secondly, it significantly reduces the evaporation of irrigation water; and thirdly, it provides additional nutrients as it decomposes.

As mentioned above, we want to integrate the branches of the lino. In the tropics, it is necessary to add slowly decomposing biomass to build up humus. Compost is converted very quickly in the soil. The lignin-containing branches seem suitable for this purpose. When the branches are cut young, the C:N ratio is still so close that decomposition does not bind nitrogen from the soil.

Mulch layer in the double row, biomass production in between (here *Canavalia*)

7.5. Manure and hydrolysate application

To significantly increase the fertilization level, two liquid fertilizers are used:

The dairy stable was converted so that deep litter manure is produced in one half and cattle manure diluted with water in the other half. This is applied directly to the double rows on a weekly basis. Fresh cattle manure is slightly acidic, which is helpful on alkaline soils. Dilution with water minimizes nitrogen losses. The nutrients from the cattle manure are also quickly available to plants.

We obtain the second liquid fertilizer from Crotalaria, which is grown on fields that are not suitable for banana production. The Crotalaria is hydrolyzed and applied either directly or via irrigation.

7.6. Outlook

The system described here is currently under development. The shade trees will only be able to take effect over the course of several years. The windbreaks will already be effective in a year. The legumes between the double rows are already giving us a wide variety of results, so that the choice of varieties, sowing density, combination of different plants, etc. can be further optimized. The effect of the two liquid fertilizers will become apparent in the coming weeks. Finally, the system will be evaluated on the basis of the harvest in January/February 2026.

8. Further results and experiences

8.1. Cultural challenges

For someone who is used to making plans, such as experimental designs, and then implementing them step by step, difficult conditions represent a major challenge. I was able to learn a lot from the participants about how to work with them. Unforeseen events always arise, and it is often impossible to predict the conditions of the next day, let alone the next week. As it is impolite to refuse something, I had to learn to distinguish between commitments that can be fulfilled in the short term and those that can only be fulfilled in the long term.

It is admirable how people deal with these situations with flexibility, a talent for improvisation and serenity.

8.2. Local expert group

The concept of a local expert group with participants from all farms has proven successful, even though the country actually has a policy of closed farm gates. No one likes to let other producers visit their farm. All participants greatly appreciated the exchange of experiences, joint observations, and discussions. Approximately 10 farm visits took place, as well as 8 meetings at which various topics were discussed in depth.

The first local coordinator was considered so competent that she was recruited by Plantaciones del Norte to head the research department. Unfortunately, it was difficult to find a second coordinator. When we did find someone, however, it became apparent after three months that she was overwhelmed by the tasks assigned to her. The project is very complex and deals with a wide range of topics at the same time.

8.3. Learning processes and motivation

The various groups of people involved have very different learning behaviors and specific motivations.

At Plantaciones, the managing directors attach great importance to scientific results. And the decisive factor is always economic efficiency.

Most practitioners are only convinced by what they see and, ultimately, only by the yields. With the workers, most of whom come from Haiti, it is advisable to teach new skills by demonstrating them and providing long-term support.

The engineers and those responsible for certification need good process documentation. There is a constant fear of doing something wrong due to a lack of knowledge (the standards are too complex and incomprehensible, and their interpretation is often perceived as arbitrary). This makes it very difficult to change anything or try something new.

For plant managers, the most important issue is the amount of work involved, as labor is becoming increasingly expensive and difficult to obtain.

8.4. Lack of basic requirements

It became clear that the farms did not have sufficient free labor capacity to undertake such extensive practical development. So we deviated from the original idea of conducting the same exploratory trials on different farms. Instead, we conducted different exploratory trials on each farm, depending on the interests and capabilities of the farms.

In Paso Robles, a dedicated team has now been formed to oversee the trials exclusively. This has significantly improved the quality of the trials.

It also became clear that there is no differentiated arable farming culture in the Dominican Republic. Very limited soil cultivation techniques (there are only disc harrows and rotary tillers, no cultivators or grubbers; there is no mechanical sowing technology for grasses and legumes; and much more) make soil preparation difficult. There is no hoeing technology for weed control. We exported a hoeing machine and introduced the first steps of the process.

Seeds are only available to a very limited extent. For this reason, we also had to set up our own seed production (Clitoria, Canavalia, Mucuna, etc.). In addition, a trial had to be completely redesigned because, after sowing, it turned out that many of the seeds were not viable.

8.5. Seed production

As shown in a trial, seeds for all desired plants are not commercially available. In particular, the important perennial legumes *Clitoria, Mucuna*, and *Canavalia* must be produced in-house. Trials have been conducted and areas are now regularly sown for seed production. Further trials are needed to determine the extent to which mechanization (combine harvesting) can be used.

8.6. Seedlings cultivation

Since Clitoria is not easy to establish in existing plantations, we have set up a nursery. Here, both legumes and trees can be grown. Flamboyant (preferred shade tree), Gliricidia, and Leucanea are grown from seeds or cuttings.

8.7. Irrigation conditions

There are three different irrigation methods available. Traditionally, flood irrigation is used for bananas, as is common for rice. This is not an optimal method for sowing. Field emergence is very irregular and seed distribution is uneven. Sowing is done by hand.

Sprinkler irrigation works best, but is only available on two farms.

This is where we achieve the best results.

The drip irrigation available on two farms is also not suitable for creating comprehensive vegetation cover. Here, we are still in the early stages of experimenting with pre-grown plants, which are then planted deep so that they can reach the moisture layer.

8.8. Joint laboratory

The intended high number of laboratory tests and the associated costs gave rise to the idea of setting up a joint laboratory. However, the low-cost offer for a nitrogen analyzer turned out to be so questionable that we ultimately decided against it. In the meantime, we have found another laboratory that offers us the tests at a lower cost. The issue of having our own laboratory will certainly be revisited, as we still see a need for it given the many questions that remain.

8.9. Networking with universities

Contacts with the University of Santiago (ISA) were promising, but unfortunately the promised literature collaboration, the performance of analyses, and the secondment of interns did not take place.

An intern from the University of Hohenheim worked on the project for three months and made a significant contribution to the establishment of the forage cultivation trial on two farms.

We recently took on an intern from the local university in Mao, who has proven to be very skilled and interested. Two of his fellow students are now interested in writing their theses on the fodder cultivation trial.

9. Conclusion and outlook

9.1 Key results and findings

As this project report makes clear, we are working on a wide range of different issues.

In some areas, we have made great strides in a short period of time:

- Crotalaria introduced as a green manure plant
- Successful small-scale hydrolysis production
- Established our own seed production (Mucuna, Canavalia, Clitoria)
- Established young plant cultivation and tree nursery
- Clarification that Utrisha does not reliably increase yields
- Willingness and initial trials with sheep grazing
- First large-scale agroforestry trial established (2.8 ha) with new cultivation system

The following further steps in these areas:

- Large-scale hydrolysis production
- Mechanized seed production, e.g., for own Crotalaria seed production
- Increase the number of tree species, find suitable seeds for the tree nursery, establish cooperation with state tree nurseries
- Expanding grazing with sheep
- Testing variants in the new agroforestry cultivation method: testing trees for suitability; developing windbreaks; testing different soil cover variants; gaining experience in building up the mulch layer, checking whether a more resilient microclimate is established

We are still in the early stages with other issues:

- Preparation and application of preparations
- Switching to manure and its application
- Development of legume-rich forage and pasture mixtures
- Feeding trials to examine the effect of new forage developments on manure quality, milk yield, and meat gain

This is an ongoing development process that will continue for several years.
In the fertilization increase trials, we observed that yields did not increase to the extent that would have been expected from the application of nitrogen quantities through so-called organic fertilizers. The reasons for this must be investigated urgently.

9.2 Further questions and their implementation

The following topics from the completed project phase should be investigated in greater depth, as they have proven to be promising options for improved nutrient management in the farm organism:

9.2.1. Forage production and grazing systems:

The experiment will provide insights into the potential of individual plant species. In a further experiment, mixtures with many individual components are now to be tested. Efforts to obtain seeds from other grasses and, above all, legumes are also continuing.

The establishment of a holistic grazing system is to be further developed with the planting of windbreaks and forage hedges. The main focus here is on biomass yields, but also on the utilization of scarce water resources.

9.2.2. Hydrolysates on a large scale and with other legumes:

The successful small-scale trials are now to be transferred to large quantities. Initial trials are currently underway. The focus here is on labor management and technical issues, as well as further optimization of seed density. To this end, a seed drill is currently being imported, which, in combination with an imported cultivator, should ensure better seed distribution and better germination conditions.

9.2.3. Agroforestry system trial:

The following topics are on the agenda here:

- Testing the trees for suitability and further diversification
- Checking the root growth of trees and shrubs with regard to competition
- Development of windbreaks: manageability of pruning; further diversification of the plants used (e.g., with Sesbania sesban)
- Testing of different soil cover variants: biomass yield; shade tolerance;
- Experience in building up the mulch layer: when is how much biomass needed for successful weed suppression, optimization of additional fertilizer application (slurry spreading)
- Checking whether a more resilient microclimate is established: temperature reduction; reduction in water consumption; development of Sigatoca infestation
- Diversification and testing of other legumes: *siratro, desmodium, lab lab*

9.2.4. Biodynamic preparations

Additional aspects that will continue to be worked on include biodynamic preparations. The well-known positive effect of horn manure preparations on root development will be investigated, as will the extent to which they improve nutrient absorption and increase yields.

- Establishment of a garden for growing plants for biodynamic preparations, including the production and storage of the preparations.
- Intensive use of silica preparation
- Demonstration experiments on root formation after the application of horn manure preparations

9.2.5. Water use

Another important aspect concerns water consumption and efficiency in its use.

9.3 Project implementation/organisation

9.3.1. Specific test planning

The test facilities and data collection for the above-mentioned issues are to be discussed and agreed upon in a workshop with the individual companies in October, provided that further funding for the project is granted.

9.3 2. Project manager on site

It has become apparent that my presence as project manager, who can offer quick solutions to questions that arise based on my experience and knowledge, has had a decisive impact on the speed of progress. I also observe more closely when I am on site.

That is why I have decided to be on site during the second phase of the project. This will also enable me to take on the role of local coordinator.

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I would particularly like to thank Jetta van den Berg for her entrepreneurial willingness to take risks, try new things, and quickly provide the necessary human resources. She enabled her employee Janne Delva to concentrate fully on this development project. Janne has completed an enormous amount of work, always reliably and with great commitment.

Karl Steiner and I were able to discuss many scientific questions in depth. He helped me a lot to come to terms with Dominican culture.

Francis Peralta has made many important contributions with scientific precision, initially as coordinator and now as the person responsible for the research and development department at Plantaciones del Norte.

I am really enjoying the project, as I am able to put my many years of agricultural experience to good use and work together with the farms to find a sustainable solution for agriculture in the Dominican Republic.